

In[*]:=

\$Assumptions = {E1 ∈ ℝ, E2 ∈ ℝ, E3 ∈ ℝ, E4 ∈ ℝ, p1 ∈ ℝ, p2 ∈ ℝ,
asunciones

p3 ∈ ℝ, p4 ∈ ℝ, E1 > 0, E2 > 0, E3 > 0, E4 > 0, p1 > 0, p2 > 0, p3 > 0, p4 > 0};

Δ[vx_, vy_, vz_] :=

Module[{v = $\sqrt{vx^2 + vy^2 + vz^2}$, γ = $\frac{1}{\sqrt{1 - (vx^2 + vy^2 + vz^2)}}$, γbar = $\frac{1}{\sqrt{1 - (vx^2 + vy^2 + vz^2)}} - 1$ },
módulo

{ {γ, -γ vx, -γ vy, -γ vz},

{-γ vx, 1 + γbar $\frac{vx^2}{v^2}$, γbar $\frac{vx vy}{v^2}$, γbar $\frac{vx vz}{v^2}$ },

{-γ vy, γbar $\frac{vx vy}{v^2}$, 1 + γbar $\frac{vy^2}{v^2}$, γbar $\frac{vy vz}{v^2}$ },

{-γ vz, γbar $\frac{vx vz}{v^2}$, γbar $\frac{vy vz}{v^2}$, 1 + γbar $\frac{vz^2}{v^2}$ }}]

P = {P1, P2, P3, P4};

{P1, P2, P3, P4} =

{ {E1, px1, py1, pz1},

{E2, px2, py2, pz2},

{E3, px3, py3, pz3},

{E4, px4, py4, pz4}}];

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In[ ]:= $Assumptions = Join[$Assumptions, { $\alpha > 0$ ,  $\beta > 0$ ,  $mB \in \mathbb{R}$ }]];
|asunciones      |junta |asunciones
{px1, px2, px3, px4} = {
  0,
  0,
  0,
  0};
{py1, py2, py3, py4} = {
  0,
  0,
  p3 Sin[ $\alpha$ ],
  |seno
  -p4 Sin[ $\beta$ ]};
|seno
{pz1, pz2, pz3, pz4} = {
  p1,
  -p2,
  -p3 Cos[ $\alpha$ ],
  |coseno
  p4 Cos[ $\beta$ ]};
|coseno
m1 = 0;
m2 = 0;
m3 = mB;
m4 = mB;

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In[*]:= E1o = E1 /. Solve[P[[1]][[1]] ==  $\sqrt{m1^2 + P[[1]][[2]]^2 + P[[1]][[3]]^2 + P[[1]][[4]]^2}$ , E1][[1]][[1]] // Normal //
|resuelve|normal

FullSimplify;
|simplifica completamente|

E2o = E2 /. Solve[P[[2]][[1]] ==  $\sqrt{m2^2 + P[[2]][[2]]^2 + P[[2]][[3]]^2 + P[[2]][[4]]^2}$ , E2][[1]][[1]] // Normal //
|resuelve|normal

FullSimplify;
|simplifica completamente|

E3o = E3 /. Solve[P[[3]][[1]] ==  $\sqrt{m3^2 + P[[3]][[2]]^2 + P[[3]][[3]]^2 + P[[3]][[4]]^2}$ , E3][[1]][[1]] // Normal //
|resuelve|normal

FullSimplify;
|simplifica completamente|

E4o = E4 /. Solve[P[[4]][[1]] ==  $\sqrt{m4^2 + P[[4]][[2]]^2 + P[[4]][[3]]^2 + P[[4]][[4]]^2}$ , E4][[1]][[1]] // Normal //
|resuelve|normal

FullSimplify;
|simplifica completamente|

p1o = p1 /. Solve[P[[1]][[1]] ==  $\sqrt{m1^2 + P[[1]][[2]]^2 + P[[1]][[3]]^2 + P[[1]][[4]]^2}$ , p1][[1]][[1]] // Normal //
|resuelve|normal

FullSimplify;
|simplifica completamente|

p2o = p2 /. Solve[P[[2]][[1]] ==  $\sqrt{m2^2 + P[[2]][[2]]^2 + P[[2]][[3]]^2 + P[[2]][[4]]^2}$ , p2][[1]][[1]] // Normal //
|resuelve|normal

FullSimplify;
|simplifica completamente|

p3o = p3 /. Solve[P[[3]][[1]] ==  $\sqrt{m3^2 + P[[3]][[2]]^2 + P[[3]][[3]]^2 + P[[3]][[4]]^2}$ , p3][[1]][[1]] // Normal //
|resuelve|normal

FullSimplify;
|simplifica completamente|

p4o = p4 /. Solve[P[[4]][[1]] ==  $\sqrt{m4^2 + P[[4]][[2]]^2 + P[[4]][[3]]^2 + P[[4]][[4]]^2}$ , p4][[1]][[1]] // Normal //
|resuelve|normal

FullSimplify;
|simplifica completamente|

In[*]:= Solve[{p1^2 == p1o^2, p2^2 == p2o^2, p3^2 == p3o^2, p4^2 == p4o^2,
|resuelve|
P[[1]][[1]] + P[[2]][[1]] == P[[3]][[1]] + P[[4]][[1]],
P[[1]][[2]] + P[[2]][[2]] == P[[3]][[2]] + P[[4]][[2]],
P[[1]][[3]] + P[[2]][[3]] == P[[3]][[3]] + P[[4]][[3]],
P[[1]][[4]] + P[[2]][[4]] == P[[3]][[4]] + P[[4]][[4]]
}, {E3, E4, p3, p4,  $\beta$ },  $\mathbb{R}$ ]

Out[*]=
$Aborted

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